Chapter 18, Problem 43P

Problem

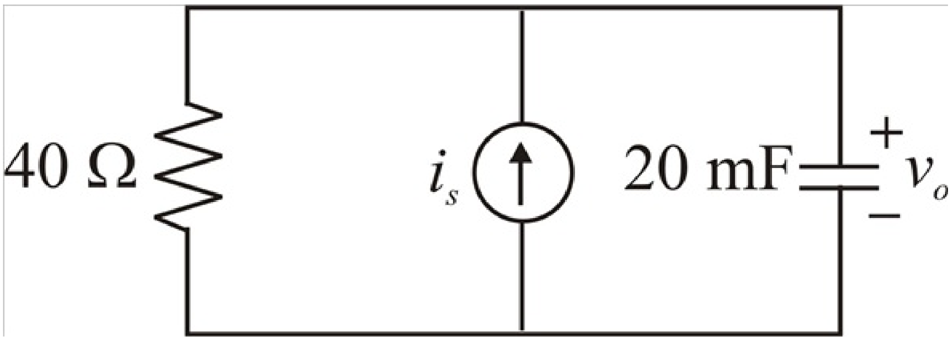
Find *vo*(*t*) in the circuit of Fig. 18.45, where *is* = 5*e*−*tu*(*t*) A.

Figure 18.45

Step-by-step solution

Step 1 of 7

Given circuit diagram is



Step 2 of 7

Given that

$$i_s = 5e^{-t}u(t)\,\text{A}$$

Applying Fourier Transform,

$$\mathcal{F}[i_s] = \mathcal{F}\left[5e^{-t}u(t)\right]$$

$$I_s(\omega) = \frac{5}{1+j\omega}\text{A}$$

Converting time domain circuit to frequency domain circuit,

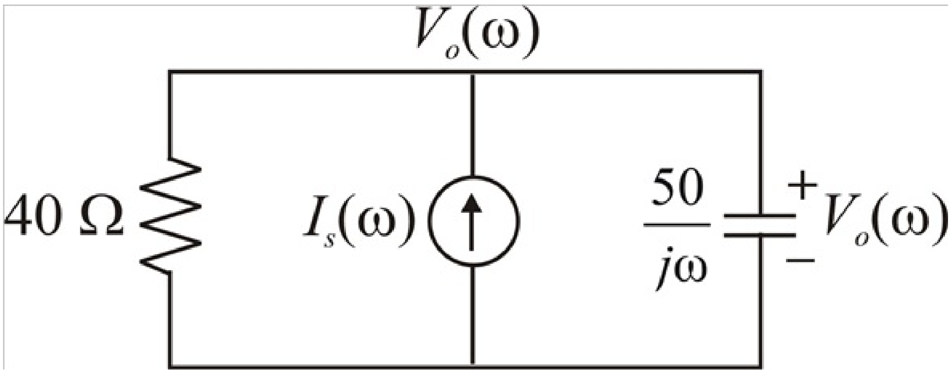
$$X_C = \frac{1}{j\omega C}$$

$$X_{20\text{mF}} = \frac{1}{j\omega \times 20\times 10^{-3}}$$

$$X_{20\text{mF}} = \frac{50}{j\omega}$$

Thus, the frequency domain circuit is

Step 3 of 7



Step 4 of 7

Applying KCL at node $V_o(\omega)$:

$$\frac{V_o(\omega)}{40}+I_s(\omega)+\frac{V_o(\omega)}{\frac{50}{j\omega}}=0$$

$$\frac{V_o(\omega)}{40}+\frac{j\omega V_o(\omega)}{50}=-I_s(\omega)$$

$$\frac{5V_o(\omega)+4j\omega V_o(\omega)}{200}=-I_s(\omega)$$

$$\frac{(5+4j\omega)V_o(\omega)}{200}=-I_s(\omega)$$

Substitute $I_s(\omega)$ in the above equation

$$\frac{(5+4j\omega)V_o(\omega)}{200}=-\left(\frac{5}{1+j\omega}\right)$$

$$(5+4j\omega)V_o(\omega)=-\left(\frac{1000}{1+j\omega}\right)$$

$$V_o(\omega)=-1000\left(\frac{1}{(1+j\omega)(5+4j\omega)}\right)$$

Step 5 of 7

Using partial fractions to above equation,

$$V_o(\omega)=\frac{1000}{(1+j\omega)(5+4j\omega)}=1000\left(\frac{A}{1+j\omega}+\frac{B}{5+4j\omega}\right) \quad \text{...... (1)}$$

$$\frac{1}{(1+j\omega)(5+4j\omega)}=\frac{A(5+4j\omega)+B(1+j\omega)}{(1+j\omega)(5+4j\omega)}$$

$$1=A(5+4j\omega)+B(1+j\omega) \quad \text{...... (2)}$$

Step 6 of 7

Put $j\omega = -1$ in equation-(2)

$$1 = A(5+4(-1)) + B(1+(-1))$$
$$A = 1$$

Put $j\omega = 0$ in equation-(2)

$$1 = A(5+4(0)) + B(1+0)$$
$$1 = 5A + B$$
$$1 = 5(1) + B$$
$$B = -4$$

Step 7 of 7

Substitute $A \text{ and } B$ in equation-(1)

$$V_o(\omega) = 1000\left(\frac{1}{1+j\omega} + \frac{-4}{5+4j\omega}\right)$$
$$V_o(\omega) = 1000\left(\frac{1}{1+j\omega} - \frac{1}{1.25+j\omega}\right)$$

Appling inverse Fourier Transform to above equation

$$\mathcal{F}^{-1}\left[V_o(\omega)\right] = \mathcal{F}^{-1}\left[1000\left(\frac{1}{1+j\omega} - \frac{1}{1.25+j\omega}\right)\right]$$
$$\mathcal{F}^{-1}\left[V_o(\omega)\right] = 1000\left(\mathcal{F}^{-1}\left[\frac{1}{1+j\omega}\right] - \mathcal{F}^{-1}\left[\frac{1}{1.25+j\omega}\right]\right)$$

Therefore, $\boxed{v_o(t) = 1000\left(e^{-1t} - e^{-1.25t}\right)\mu(t)\ \text{V}}$.